Galaxy Spotter
DATA11001 Group project report — October 2026
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1 Introduction

For our group project, we have built an application that lets, whether it be hobby or professional, astronomers identify the type of galaxy from an image. Galaxies can generally be considered spirals or ellipticals based on their structure, or "morphology". Ellipticals have a more Gaussian distribution of matter, whereas in spirals the stars and interstellar matter collect into so-called spiral arms. Spirals and ellipticals have different ages, chemical make-ups and stellar populations, making it important for the astronomer to know which type they are working with.

One main challenge in classification arises from data quality and processing; the correct transformations have to be applied to the raw images for the object to be clearly visible. Even then, especially when the object is dim or small in the frame, it can be difficult to make the right call. This is only amplified when working with large datasets.

To counter this limitation, we have trained a Convolutional Neural Network model on 10,000 real, labelled observational images of galaxies from Legacy Survey Data Release 11 (Dey et al., 2019), reaching final accuracies of above 80%. This report outlines the process of building the application, and is structured as follows: Section 2 explains the collection and processing of training data, Section 3 covers the CNN model we built and expands on this process, Section 4 focuses on the user interface and incorporating the model into a usable application, and finally, Section 5 summarizes our work and offers reflections on the project as a whole.

2 Data

2.1 Sourcing

2.1.1 Galaxy Zoo 2

For our classifier, we wanted a large sample of galaxies that were of roughly equal angular size. The Galaxy Zoo 2 project (GZ2) (Hart et al., 2016) was the perfect source for such a data set. This is designed around citizen science volunteers classifying galaxy morphology of angularly large galaxies and provides a table of the coordinates and classification labels for nearly $300,000$ galaxies. For our project, we used a subsample of $10,000$ galaxies that gave use a roughly $1:1$ ratio of spiral and elliptical galaxies.

2.1.2 Legacy Survey

However, GZ2 does not easily provide the actual image files of the galaxies. To acquire the images, we used the Legacy Survey cutout service. This is a web link provided by the Legacy Survey that produces a 256×256 pixel image based on input parameters. This link is of the form:

www.legacysurvey.org/viewer/fits-cutout? ...
ra=RA&dec=DEC&layer=LAYER&pixscale=PIXSCALE&bands=BAND

where RA and DEC are the right ascension and declination, respectively, in degrees; PIXSCALE is the pixel scale in arcmin/pixel; LAYER is the image layer from which to pull the image, for which we used ls-dr11 (Legacy Survey Data Release 11, Dey et al., 2019); and BAND is the wavelength filter band (typically r, g, or z; for our training, we used r-band). Data Release 11 was chosen because, after testing other options, it provided the highest sky coverage, ensuring that all training galaxies contain images.

Using the list of $10,000$ galaxy coordinates from GZ2 and the Legacy Survey cutout service, we collected a FITS image for each galaxy. We used FITS images, as they contain the exact pixel values of the galaxy without compression or interpolation.

2.2 Preprocessing

To prepare the images for training, we needed to do a number of steps to clean up the images. An example of these can be seen in Figure 1.

2.2.1 Background Subtraction

The images naturally contain background noise that could interfere with the classification. We used the `python` package `sep` to extract the image background and remove it from the image. Visually, this makes the foreground objects stand out more and reduces any noise within the background.

2.2.2 Subject Isolation

Since we only require the main galaxy within the image for our classification, we needed to remove all other objects from the frame. As part of the background subjection procedure, `sep` also provides a list of all identified objects with an encompassing ellipse. With this list,

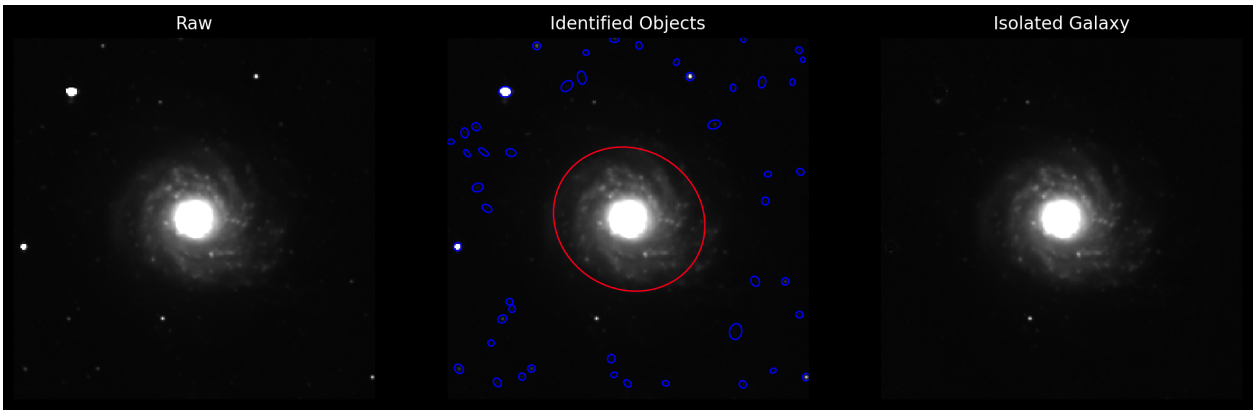


Figure 1: This is an example of the preprocessing pipeline applied to an image of a spiral galaxy. The left panel is the raw image from the Legacy Survey Cutout service. The center panel is background extracted and has the objects identified. The red ellipse is the identified target galaxy and the blue ellipses are the identified non-subject objects. The right panel is the final image with the target galaxy isolated.

we used `sep.mask_ellipse` to replace each non-subject object with RMS background noise, leaving only our target galaxy within each image. However, we didn't want to remove any of the target galaxy, so if the ellipse of a non-subject object overlapped the target galaxy, it was not removed. There are some artifacts remaining in a few of the training images, but this had to be left in as an irreducible minimum.

We also checked how close all of our galaxies are to each other. If an ellipse of one galaxy was present in the image of another, both galaxies were thrown out. This removed overlapping field galaxies and merging galaxies and ensured that each image had exactly one focus galaxy.

2.2.3 Image Stretch

Finally, the cleaned image was stretched using an inverse hyperbolic sine function (*arcsinh*). This is a common practice when working with astronomical images as it brightens up the subject and minimizes the background. The process of stretching the images also normalized the pixel values to $[0, 1]$ which was required for training.

3 Convolutional Neural Network

The Convolutional Neural Network (CNN) (Lecun et al., 1998) was implemented using PyTorch and consists of four convolutional blocks followed by a fully connected classifier. Each block contains a 3×3 convolution, batch normalisation, ReLU activation, and 2×2 max-pooling. The number of feature channels increases from 32 to 64, 128, and 256, allowing the model to learn increasingly complex features from grayscale galaxy images.

An example of one convolutional block is shown below:

```
nn.Conv2d(32, 64, kernel_size=3, padding=1),
nn.BatchNorm2d(64),
nn.ReLU(inplace=True),
nn.MaxPool2d(2, 2),
```

After feature extraction, adaptive average pooling reduces the feature maps to a 256-dimensional vector. The classification head consists of two linear layers with 64 hidden neurons and dropout rates of 0.5 and 0.3 to reduce overfitting.

The model was trained using the AdamW optimizer with a weight decay of 0.0001 and binary cross-entropy loss with logits (BCEWithLogitsLoss), which is suitable for binary classification tasks.

3.1 Training and Validation

The dataset was divided into training, validation, and test sets using stratified sampling to preserve the proportion of elliptical and spiral galaxies. After each epoch, the model was evaluated on the validation set to monitor its performance.

Model performance was measured using accuracy and balanced accuracy. While accuracy represents the proportion of correctly classified galaxies, balanced accuracy gives equal importance to both galaxy classes:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

$$\text{Balanced Accuracy} = \frac{\text{Elliptical Accuracy} + \text{Spiral Accuracy}}{2}$$

To prevent overfitting, early stopping was implemented based on validation loss. The model was saved whenever validation loss improved, and training was stopped if no improvement occurred for a predefined number of consecutive epochs. Finally, the best saved model was evaluated on the independent test set to assess its generalisation performance. This model reached a balanced accuracy of 83.97%.

4 User Interface

We decided the most useful deliverable for the end user to be an application, with the functionality to import FITS images or input the coordinates of the desired object, and processing is done automatically. The application was built using PyQt6 ("PyQt6 Tutorial 2026, Create Python GUIs with Qt", 2026), keeping the configuration as simple and easy to use as possible, including buttons to import and save the images, and a breakdown of the classification confidence on a panel on the right. All are compiled with the library's built-in widgets, layouts, and dialogues. The overarching aim was to deliver an intuitive product that is applicable for both hobby astronomers and professionals. Figure 2 explains the application layout in more detail.

We chose to make the application available on github (https://github.com/pf1cha/galaxy_spotter), and prepared a complete `README` with instructions for the user and additional details on the application and model.

5 Reflection and conclusions

Multiple phases of trial-and-error were carried out during the process. Guided by the goal and data source we determined at the beginning of the project, we were able to reach the

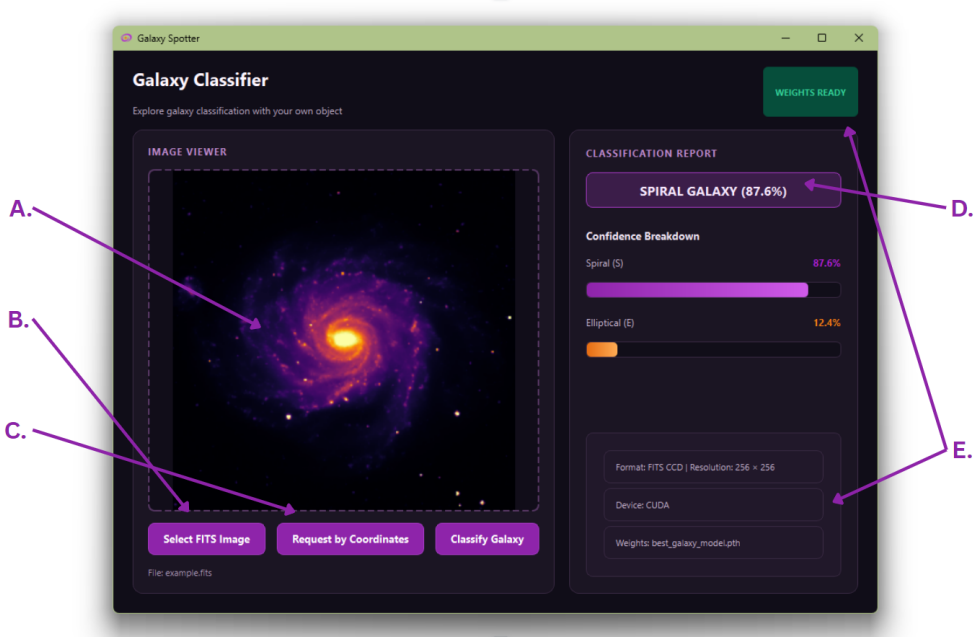


Figure 2: A screenshot of the application after running classification on an image. A. The processed image displayed for the user. B. Button to select FITS image from the system. C. Opens a window to input coordinates of the object, which the application then pulls online. D. Classification report, outlining the results and a confidence breakdown, which is important for researchers with large datasets. E. Image and device information, and the file defining model parameters. If there is no valid file, the top icon will inform the user. Example before image input is shown in Appendix 1.

preprocessing stage smoothly. At this stage, however, we had to try multiple methods to consistently clean FITS images for thousands of galaxies. A few galaxies still returned completely empty images, which we had to identify and clean out before training the model.

Model training was done by all group members, testing different configurations (number of epochs, learning rate, batch size), until the final model was obtained. One challenge involved the number of epochs, which would influence model. A high number of epochs would often involve the training stop, but often resulted in a skewed model with significantly higher accuracy for one of the classes, neglecting the other. A moderate number of epochs (5), generally yielded a more balanced model. The final model used was remarkably balanced, with 85.45% elliptical accuracy and 82.50% spiral accuracy.

One limitation observed was a potential bias related to galaxy size: smaller galaxies were more often classified as elliptical, while larger ones were more often classified as spiral. This may reflect the training data, as elliptical galaxies tend to appear smaller because they are further away. A possible improvement would be to normalise galaxy sizes by adjusting image cutouts to a standard size, but this was beyond the scope of the project.

Future work could improve classification accuracy through hyperparameter tuning, experimenting with different numbers of convolutional layers, and implementing learning rate scheduling. Further investigation of dataset biases and improved image preprocessing could also help the model generalise better to different galaxy sizes. Additionally, collecting more diverse galaxy images and expanding the classification task to include more galaxy types would make the model more versatile.

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Appendix

Appendix 1: User Interface before input

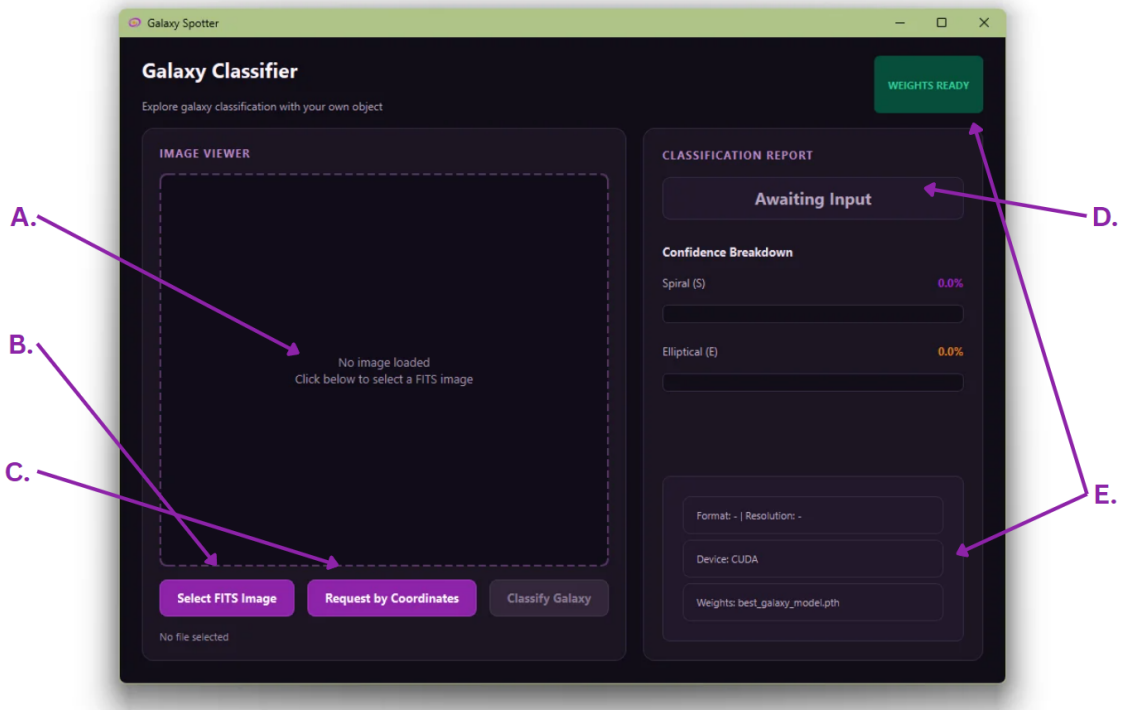


Figure 3: A screenshot of the application before an image input is given. A. Instructing the user to load an image. B. To select FITS image from system. C. Opens a window to input coordinates of the object, which the application then pulls online. D. Classification report awaiting input. E. Device information, and the file defining model parameters. If there is no valid file, the top icon will inform the user.